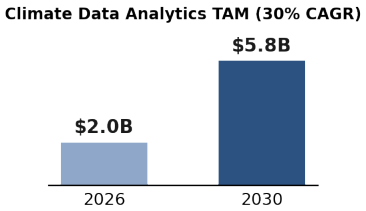
AI-Native Climate and Weather Intelligence

Opportunity

We are living through simultaneous AI and climate transformation. The same compute driving up energy demand is also collapsing the two constraints that have bottlenecked physics-based climate modeling for decades: speed and accuracy. NVIDIA's Earth-2 platform (Jan 2026) runs AI weather emulation up to 1,000x faster than physics simulation; Google's WeatherNext 3 delivers 5x sharper resolution with hourly satellite updates. In September 2026, an unreleased OpenAI model coordinating 10,000 agents produced a disputed but significant proof touching the Navier-Stokes equations - the equations underlying every hydrodynamic and atmospheric model in use, including the ones I worked with throughout my career. Whether or not that specific result holds up, the direction is clear: AI is evolving fast to probing the equations and algorithm underneath existing physics models directly, creating an early venture category - AI-native climate and weather intelligence - that I genuinely believe is investable. The addressable market for climate data analytics is estimated at $2.0B in 2026 growing to $5.8B by 2030 at a 30% CAGR, among the fastest growing subsegments in climate tech.



Investment Criteria

I would evaluate startups in this space against three criteria:

- Proprietary data or sensing advantage, not a thin wrapper on open foundation models. The lesson from physical climate risk investing (One Concern, First Street, Fathom) is that reselling insight without a defensible data or distribution moat gets commoditized or acquired for parts.
- Recurring, contracted revenue from a vertical with real switching cost, measured by ARR growth and the share of revenue in multi-year contracts rather than pilots.
- Deep technical domain expertise on the founding team. Fine-tuning foundation models for a vertical (flood extent, wildfire spread etc.) requires physical-science fluency most generalist AI teams don't have.

Given how early this category is, I'd focus on seed through Series B, where teams have proven technical differentiation but haven't yet had to defend it against incumbents at scale. Three companies I'm tracking:

Company	Proprietary Edge	Proof Point
Tomorrow.io	Own satellite constellation + in-house AI model (CBAM)	~$100M self-reported ARR, $1B+ valuation (Feb 2026)
WindBorne Systems	Own weather balloon network + own AI model (WeatherMesh-6)	$250M valuation, revenue tripled YoY
Reask	Physics-based cyclone modeling, proprietary windfield data	Embedded in Swiss Re & Kettle underwriting

Risks

- Political risk: federal climate and AI policy remain volatile and unsettled. a hostile turn on either front could compress public-sector demand and private capital availability at once.
- AI explainability: insurers and regulators require auditable, defensible risk models to satisfy capital-adequacy requirements - which is why RMS and Verisk have survived for decades. Black-box AI models may face real adoption friction in the exact vertical (insurance) most likely to pay premium prices.
- Platform risk: the risk that value captured by a startup gets absorbed by a larger platform it depends on. A company built as a thin layer on top of a freely available foundation model has no defensible moat unless it owns proprietary data or distribution that NVIDIA and Google don't have.

Summary

AI is collapsing the compute speed and accuracy constraints that have bottlenecked physical-world modeling for a century, creating an early-stage venture category in climate and weather intelligence. The winners will be teams with proprietary data or deep domain expertise, not thin wrappers on someone else's foundation model.